

BACS - HW (Week 10)

Let's go back and take another look at our analysis of the cars dataset. Recall our variables:

1. mpg: miles-per-gallon (dependent variable)
2. cylinders: cylinders in engine
3. displacement: size of engine
4. horsepower: power of engine
5. weight: weight of car
6. acceleration: acceleration ability of car (seconds to achieve 0-60mph)
7. model_year: year model was released
8. origin: place car was designed (1: USA, 2: Europe, 3: Japan)

Did you notice the following from doing a full regression model of mpg on all independent variables?

- Only displacement, weight, year, and origin had significant effects
- The factors cylinders, displacement, and horsepower were highly correlated with weight
- Displacement even had the opposite effect in the regression from its visualized effect!
- Some factors, like horsepower, seem to have a nonlinear (exponential) relationship with mpg

Question 1) Let's deal with **nonlinearity** first. Create a new dataset that log-transforms several variables from our original dataset (called cars in this case):

```
cars_log <- with(cars, data.frame(log(mpg), log(cylinders), log(displacement),  
                                log(horsepower), log(weight), log(acceleration),  
                                model_year, origin))
```

Note: unless you specify column names, each log transformed column <col> is named: <col>. Log.

- a. Run a new regression on the cars_log dataset, with mpg.log. dependent on all other variables
 - i. Which log-transformed factors have a significant effect on log.mpg. at 10% significance?
 - ii. Do some new factors now have effects on mpg, and why might this be?
 - iii. Which factors still have insignificant or opposite (from correlation) effects on mpg?
Why might this be?
- b. Let's take a closer look at weight, because it seems to be a major explanation of mpg
 - i. Create a regression (call it regr_wt) of mpg over weight from the original cars dataset
 - ii. Create a regression (call it regr_wt_log) of log.mpg. on log.weight. from cars_log
 - iii. Visualize the residuals of both regression models (raw and log-transformed):
 1. density plots of residuals
 2. scatterplot of log.weight. vs. residuals
 - iv. Which regression produces better distributed residuals for the assumptions of regression?
 - v. How would you interpret the slope of log.weight. vs log.mpg. in simple words?
 - vi. From its standard error, what is the 95% confidence interval of the slope of log.weight. vs log.mpg.?

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Question 2) Let's tackle **multicollinearity** next. Consider the regression model:

```
regr_log <- lm(log.mpg. ~ log.cylinders. + log.displacement. + log.horsepower. +  
               log.weight. + log.acceleration. + model_year +  
               factor(origin), data=cars_log)
```

- a. Using regression and R^2 , compute the VIF of `log.weight.` using the approach shown in class
- b. Let's try a procedure called *Stepwise VIF Selection* to remove highly collinear predictors.
Start by Installing the 'car' package in RStudio -- it has a function called `vif()`
(note: CAR package stands for "Companion to Applied Regression" -- it isn't about cars!)
 - i. Use `vif(regr_log)` to compute VIF of the all the independent variables
 - ii. Eliminate from your model the single independent variable with the *largest* VIF score that is also greater than 5
 - iii. Repeat steps (i) and (ii) until no more independent variables have VIF scores above 5
 - iv. Report the final regression model and its summary statistics
- c. Using stepwise VIF selection, have we lost any variables that were previously significant?
If so, how much did we hurt our explanation by dropping those variables? (hint: look at model fit)
- d. From only the *formula* for VIF, try deducing/deriving the following:
 - i. If an independent variable has no correlation with other independent variables, what would its VIF score be?
 - ii. Given a regression with only two independent variables (X_1 and X_2), how correlated would X_1 and X_2 have to be, to get VIF scores of 5 or higher? To get VIF scores of 10 or higher?

Question 3) Might the relationship of weight on mpg be different for cars from different origins?

Let's try visualizing this. First, plot all the weights, using different colors and symbols for the three origins:
(you may choose any three colors you wish or plot this using ggplot etc. – the code below is for reference)

```
origin_colors = c("blue", "darkgreen", "red")  
with(cars_log, plot(log.weight., log.mpg., pch=origin, col=origin_colors[origin]))
```

- a. Let's add three separate regression lines on the scatterplot, one for each of the origins.
Here's one for the US to get you started:

```
cars_us <- subset(cars_log, origin==1)  
wt_regr_us <- lm(log.mpg. ~ log.weight., data=cars_us)  
abline(wt_regr_us, col=origin_colors[1], lwd=2)
```

- b. *[not graded]* Do cars from different origins appear to have different weight vs. mpg relationships?

We will investigate these relationships more in class!